

Multiple Choice (10 marks)

1. The first number in a number pattern is 28. The pattern rule is to add 14 to get the next number in the pattern. If the pattern continues, which statement is true?
 - (a) All the numbers in the pattern can be divided equally by 10.
 - (b) All the numbers in the pattern can be divided equally by 4.
 - (c) All the numbers in the pattern can be divided equally by 8.
 - (d) All the numbers in the pattern can be divided equally by 7.
2. Solve the equation $14 = w + 23$.
 - (a) -37
 - (b) -9
 - (c) 9
 - (d) 37
3. Erin made 12 pints of juice. She drinks 3 cups of juice each day. How many days will Erin take to drink all of the juice she made?
 - (a) 2 days
 - (b) 4 days
 - (c) 8 days
 - (d) 9 days
4. Mr. Jones rolls a six-sided cube numbered 1, 2, 3, 4, 5, 6. What is the probability he rolls a three?
 - (a) 6-Jan
 - (b) 5-Jan
 - (c) 3-Jan
 - (d) 2-Jan
5. If $f(x) = 2$ for all real numbers x , what is the value of $f(x + 2)$?
 - (a) 2
 - (b) 4
 - (c) 0
 - (d) 1

True / False (10 marks)

Answer True or False

6. True or False: The sum of 365 and 56 is 421.
7. True or False: The number eighteen thousand ninety-six can be expressed in expanded form as $10,000 + 8,000 + 90 + 6$.
8. True or False: The inequality $x^2 - 6x + c < 0$ has real solutions for c in the interval $(0, 9)$.
9. True or False: The expression $-105 + (-14) + 34$ simplifies to -85 .
10. True or False: The composition $f(g(h(x)))$ results in the expression $15x^2 + 15$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Determine the equation of the tangent line to the curve defined by $F(t) = (t^2 + 1, 2^t)$ at the point where $y = 4$. Explain your reasoning and calculations in detail.
12. Explain the concept of a P-value in hypothesis testing. How does it help determine the validity of the null hypothesis, and why is it important in statistical analysis?

End of Paper